TUTORIAL – 12
AI-Powered Software Testing
Aim:
To analyze AI-powered software testing tools and apply Generative AI techniques for test case generation, test data creation, defect prediction, requirement analysis and automated Selenium test script generation for software quality assurance.

SOURCE CODE :

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

driver = webdriver.Chrome()
driver.maximize_window()

try:
    # 1. Open Website
    driver.get("https://www.saucedemo.com/")

    wait = WebDriverWait(driver, 10)

    # 2. Find Login Elements
    username = wait.until(
        EC.visibility_of_element_located(
            (By.ID, "user-name")
        )
    )
```

```
    password = driver.find_element(
        By.ID, "password"
    )

    login_button = driver.find_element(
        By.ID, "login-button"
    )

    # 3. Enter Login Details
    username.send_keys("standard_user")
    password.send_keys("secret_sauce")
    login_button.click()

    # 4. Validate Products Page
    wait.until(
        EC.url_contains("inventory")
    )

    assert "inventory" in driver.current_url

    print("AI-Assisted Selenium Test: PASS")
    print("Requirement Validation: PASS")

except AssertionError:
    print("AI-Assisted Selenium Test: FAIL")

finally:
    driver.quit()

print("Test Execution Completed")
```

OUTPUT SCREENSHOT :



